Project Udyam Digital — Comprehensive Technical & Business Submission

Airtel SYNC 2026 · Enterprise Track A (Digital Journey) · Operator: Accel Business

GitHub Code Repository: https://github.com/anshraz27/Airtel_Sync.git (Branch: main)

1. Executive Summary & Market Problem

Small businesses in India (6 Crore+ MSMEs) increasingly require fixed broadband, business SIMs, WhatsApp channels, cloud backup, and security tools. However, Accel Business loses **92.6% of SMB leads** (smb_leads.csv). The steepest drop occurs at the **Quote stage (only 44% conversion)** due to complex codes and price opacity. Acquisition cost is ■15,000 per activated customer with a 17.6-day turnaround and 3.0 follow-up chase calls. Project Udyam Digital replaces this paper-heavy journey with a self-serve mobile flow that activates small businesses in one sitting under 5 minutes.

2. Target Personas & Curated Bundles (Key Question 1)

Priority & Share	Persona & Needs	Curated Bundle	Monthly Price & Unit Economic
Tier 1 (55% Base) High Volume	Micro Retail (Kirana, Salon) UPI WiFi, WhatsApp Store	Biz Fibre 100 (P01) + Secure WiFi (P22) + Postpaid (P07)	■1,947/mo Setup: ■1,000 (Waived on 24mo)
Tier 2 (25% Base) High ARPU	Professional Offices (Clinic, CA) 99.9% uptime, Cloud Email, Backup	Biz Fibre 300 (P02) + Email (P16) + Endpoint Sec (P20) + Failover (P09)	■4,813/mo Consolidated single bill
Tier 3 (15% Base) High LTV	Small Multi-Site (Restaurants) SD-WAN, CCTV, multi-site SLA	Biz Fibre 300 (P02) + SD-WAN (P10) + Cloud CCTV (P23) + Failover (P09)	■7,200/mo Centralized network control
Tier 4 (5% Base) Specialist	Small Manufacturers (Textile, Auto) 1:1 Leased Line (ILL), Machine IoT	ILL 20 (P05) + Machine IoT (P24) + Endpoint Security (P20)	■12,000/mo Guaranteed enterprise SLA

3. Conversational AI Advisor & Architecture (Key Question 2)

- **Free Hugging Face Stack:** Runs on *Qwen/Qwen2.5-72B-Instruct* or *meta-llama/Llama-3.3-70B-Instruct* via Serverless Inference API (Free tier). Evaluated alternatives: DeepSeek-R1-Distill-Qwen-32B, Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3.
- **Vernacular Multilingual:** Fluidly supports English and Devanagari Hindi (e.g. '■■■■ ■■■■ ■■■ ■■■■■■ ■■■■■ ■■, WiFi ■■ UPI ■■■■■').
- **Deterministic Eligibility Guard:** Product rules (city tier, employee bounds, sites, GSTIN requirement, dependency chain) are enforced in pure Python (app/eligibility.py). The LLM cannot hallucinate ineligible bundles.
- **100% Offline Demo Resilience:** Built-in rule-based fallback advisor activates if Hugging Face rate-limits, guaranteeing zero demo failure.

4. Smart KYC Pipeline & Planted Trap Defenses (Key Question 3)

Our Smart KYC pipeline (app/kyc.py) was tested against all 12 specimen applicant document sets (K01–K12) and caught 100% of planted traps:

1. **K01:** Flagged blank Udyam registration number + Cancelled GSTIN status in mock registry.
2. **K02:** Detected rotated canvas (1446x1894); applied automatic orientation correction.
3. **K04:** Edge-variance Laplacian sharpness score was 175.8 (< 300 threshold); prompted for clear re-upload.
4. **K05:** Caught entity name mismatch between GST Trade Name (Balaji Dental Care LLP) and Electricity Bill (Balaji Realty LLP).
5. **Freshness Check:** Enforced 90-day address proof validity against reference date 2026-07-15.
6. **DPDP Act 2023:** Pre-KYC explicit consent gate, PII masking across logs/UI (37IHK*****L1ZS), and 30-day automated file shredding.

5. Partner Synergy & Target Business Metrics (Key Questions 4 & 5)

• **ML Propensity Model:** Trained Scikit-Learn Logistic Regression on 5,000 historical leads (ROC-AUC: 0.691). Scores inbound leads and prioritizes high-intent prospects for priority partner outreach.
• **Warm Lead Recovery Queue:** Digital drop-offs at Quote/Needs stages are routed to local channel partners with pre-filled context and customized discount hooks, preserving sales commissions.
• **Target Metrics:** Lead-to-Activation conversion lifted from 7.4% to ≥**15.0% (2x)**; CAC slashed from ■15,000 to ≤**■5,000 (3x lower)**; activation cycle compressed from 17.6 days to ≤**5-7 days**; status chase calls reduced from 3.0 to **<0.2**.
• **Financial ROI:** Generates ■15 Crore in quarterly operational savings for Accel Business against <■25,000 in serverless API costs (>6,000x ROI).

6. Code Repository & Submission Deliverables

- **GitHub Repository (Main Branch):** https://github.com/anshraz27/Airtel_Sync.git
- **Working Prototype Web Portal:** FastAPI backend + Single Page App running at http://localhost:8000
- **7-Slide Submission Presentation:** Airtel_SYNC_Udyam_Digital_7Slides.pptx (PPT format) and Airtel_SYNC_Udyam_Digital_7Slides.pdf (PDF format)
- **DPDP Act Privacy Note:** privacy_note.md (Full compliance documentation)